EXPERIMENTAL STUDY OF THE LLC-TYPE SERIES RESONANT CONVERTER

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ABSTRACT

Previous studies of the LLC-type series resonant converter (LLC-SRC) have shown that this converter possesses much better control characteristics and lower component stresses compared with those of the conventional SRC. In this paper, we extend our previous work on the LLC-SRC by presenting a complete set of design curves and formulating a systematic design procedure for the converter. Based on the design procedure, a 150 W, 200 kHz, multi-output power supply unit has been built to verify our theoretical results.

I. INTRODUCTION

The conventional series resonant converter (SRC) possesses the properties of zero-current-switching, low converter voltage gain, and incorporating the leakage inductance of the output transformer in the converter design. These are desirable features for high frequency and low output voltage applications. However, the output voltage regulation of the SRC is very poor because a very wide range of the switching frequency is required for both the load and line regulations [1, 2]. To overcome this difficulty, we may use the PWM or phase-shift control and operate the SRC at its resonant frequency, resulting in the increase of the circuit complexity and the design difficulties [3, 4].

We have shown in our previous work [5] that the aforementioned difficulty of the SRC can be alleviated by connecting an additional inductor in parallel with the resonant capacitor of the conventional SRC, resulting in a new converter topology named the LLC-SRC. The idealized half-bridge converter circuit is shown in Fig. 1. It was shown in [5] that the control characteristics of the new converter is much improved as compared to those of the SRC while the advantages of the SRC are still maintained. In this paper, we expand our previous work on the LLC-SRC by presenting a complete set of design curves and formulating a systematical design procedure for the converter. Based on the design procedure, a 150 W, 200 kHz, multi-output prototype converter has been deigned and tested to verify our theoretical work. Experimental results under various load and line conditions and the comparison to their theoretical values are also presented.

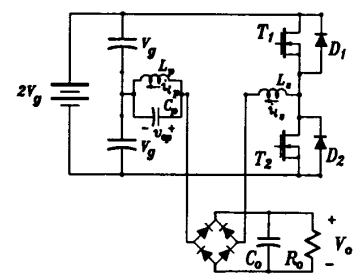

Fig. 1 The LLC-type series resonant converter

II. STEADY STATE ANALYSIS

For the circuit of Fig. 1, we assume that under the continuous conduction mode of operation, the switching devices *T1/D1* and *T2/D2* are ideal and *T1* and *T2* are operated at 50% duty cycle over a switching period T_s. Thus we may represent the effect of the switching devices by means of an equivalent square wave voltage source, $v_i(t)$, with amplitudes $\pm V_g$ and period T_s. Furthermore, we assume that C_o is large enough so that the output rectifying circuit can be replaced by a dependent voltage sink $v_E(t)$, represented by

$$v_E(t) = \frac{i_{ls}(t)}{|i_{ls}(t)|} V_o \tag{1}$$

with voltage polarity depending upon the sign of $i_{ls}(t)$. The equivalent circuit of the LLC-SRC of Fig. 1 is shown in Fig. 2. Depending on the values of $v_E(t)$ and $v_s(t)$, the equivalent circuit possesses four linear circuit modes over the switching period as shown in Fig. 3. Table I shows the conduction states of each switching device in each circuit mode and the conditions for mode change.

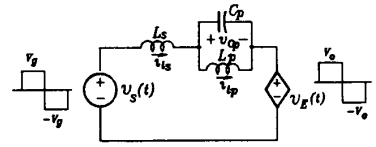

Fig. 2 The equivalent circuit of Fig. 1

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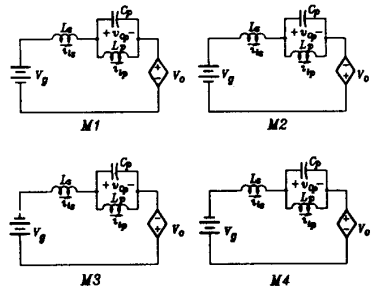


Fig. 3 The four linear circuit modes

It can be seen from Fig. 2 that C_p and L_p introduce a transmission zero at frequency $\omega_{op} = 1/\sqrt{L_p C_p}$. The natural resonant frequency of the equivalent converter circuit, which is also the transmission pole, occurs at ω_o with $\omega_{op} < \omega_o$, where $\omega_o = 1/\sqrt{1/LC_p}$ and $L = (L_s L_p)/(L_s + L_p)$. By designing ω_{op} closer to ω_o, the transmission characteristics of the circuit between ω_{op} and ω_o become sharper, resulting in a more abrupt control characteristics of the converter. Thus, the control characteristics of the converter can be improved by properly choosing the ratio of ω_{op}/ω_o or L_s/L_p.

TABLE I

Mode	Switches on	Sign of the Rectifier Current	$v_s(t)$	$v_E(t)$
M1	*T1*	$i_{ls}(t) > 0$	$+V_g$	$+V_o$
M2	*T1/D1*	$i_{ls}(t) < 0$	$+V_g$	$-V_o$
M3	*T2*	$i_{ls}(t) < 0$	$-V_g$	$-V_o$
M4	*T2/D2*	$i_{ls}(t) > 0$	$-V_g$	$+V_o$

The steady state response of the converter can be derived from the equivalent circuit of Fig. 2. The state equations for the equivalent circuit are given below.

$$\frac{d}{dt}\begin{bmatrix} i_{ls} \\ i_{lp} \\ v_{cp} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -\frac{1}{L_s} \\ 0 & 0 & \frac{1}{L_s} \\ \frac{1}{C_p} & -\frac{1}{C_p} & 0 \end{bmatrix}\begin{bmatrix} i_{ls} \\ i_{lp} \\ v_{cp} \end{bmatrix} + \begin{bmatrix} \frac{1}{L_s} & -\frac{1}{L_s} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} v_s(t) \\ v_E(t) \end{bmatrix} \tag{2}$$

Equation (2) represents a third-order system with a nonlinear voltage sink, which is difficult to solve by the conventional circuit approach. However, it can be shown that, by using the following state transformation

$$\begin{bmatrix} i_l \\ i_p \\ v_c \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} i_{ls} \\ i_{lp} \\ v_{cp} \end{bmatrix} \tag{3}$$

the new state variables v_c and i_l can be decouple from i_p so that their steady state response can be represented by a two-dimensional state-plane diagram. Substituting eq. (3) into eq. (2), the state equations in terms of i_l and v_c can be obtained.

$$\frac{di_l}{dt} = \frac{1}{L}\left[[1-(\frac{\omega_{op}}{\omega_o})^2][v_s(t)-v_E(t)]-v_c\right] \tag{4}$$

$$\frac{dv_c}{dt} = \frac{1}{C_p} i_l \tag{5}$$

With all voltages and currents normalized by V_g and V_g/Z_o, respectively, the state-plane equation is given by

$$\frac{dv_{nc}}{di_{nl}} = \frac{i_{nl}}{v_{ns}-v_{nE}-v_{nc}} \tag{6}$$

where

$$v_{ns}=[1-(\frac{\omega_{op}}{\omega_o})^2]\frac{v_s(t)}{V_g} \quad and \quad v_{nE}=[1-(\frac{\omega_{op}}{\omega_o})^2]\frac{v_E(t)}{V_g} \tag{7}$$

When the converter is operated in the continuous conduction mode, the steady state solution of eq. (6) consists of four circular arcs in the $v_{nc}-i_{nl}$ plane over a switching period T_s, which are described by the following equation.

$$i_{nl}^2+(v_{nsi}-v_{nEi}-v_{nc})^2 = V_{nmi}^2 \quad for\ t_i-1 \le t \le t_i \tag{8}$$

where $i = 1, 2, 3, 4$ *and* V_{nmi} are the radii which can be determined from the boundary conditions. The centers of these circular arcs depend on the value of $v_s(t)$ and the sign of $i_{ls}(t)$ and are located at $(V'_{ng}-V'_{no}, 0)$, $(V'_{ng}+V'_{no}, 0)$, $(-V_{ng}+V_{no}, 0)$, and $(-V'_{ng}-V'_{no}, 0)$, where $V'_{ng}=1-(\omega_{op}/\omega_o)^2$, $V'_{no}=[1-(\omega_{op}/\omega_o)^2]M$, and $M = V_o/V_g$ which is the converter voltage gain. A typical state-plane diagram of the converter is given in Fig. 4.

Based on the state plane diagram of Fig. 4, we derive the following expressions for the normalized average output and input currents

$$I_{no}=\frac{2}{T_s}\int_{t_o}^{t_o+T_s/2} |i_{nls}(t)|\,dt=\frac{2}{\gamma}[1-(\frac{\omega_{op}}{\omega_o})^2][v_{nc}(t_1)-\alpha\beta(\frac{\omega_{op}}{\omega_o})^2] \tag{9}$$

$$I_{ng} = \int_{t_o}^{t_o+T_s/2} i_{nls}(t)dt = -\frac{2}{\gamma}[1-(\frac{\omega_{op}}{\omega_o})^2][v_{nc}(t_o)+\alpha\beta M(\frac{\omega_{op}}{\omega_o})^2] \quad (10)$$

Where $\gamma = \omega_o(T_s/2) = \pi(f_o/f_s) = \pi/f_{ns} = \beta+\alpha$. β and α are the conduction angles of $T1$ and $D1$, respectively. Since the converter is assumed lossless, conservation of power, $I_oV_o = I_gV_g$, leads to the following relation

$$M = -\frac{v_{nc}(t_o)}{v_{nc}(t_1)} \quad (11)$$

Thus, the complete response of the converter as well as I_{ng}, I_{ng}, and M can be determined from the above equations once the state-plane diagram is obtained.

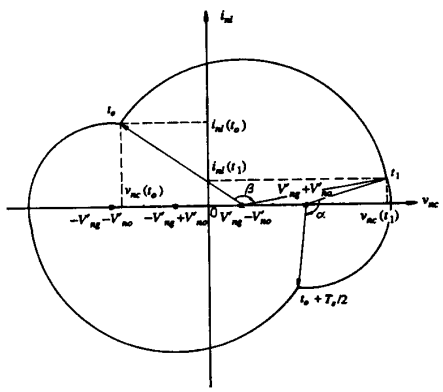


Fig. 4 Typical state-plane diagram of the LLC-type SRC

III. CHARACTERISTIC CURVES

From the design point-view, the parameters ω_{op}/ω_o, M, γ, and Q_s, where $Q_s = Z_s/R_o$, are design parameters. The normalized boundary voltages and currents $v_{nc}(t_o)$, $i_{nl}(t_o)$, $v_{nc}(t_1)$, and $i_{nl}(t_1)$ are state-plane parameters because the state-plane diagram can be characterized by these parameters. In this section, we will show that based on a set of equations derived from the state-plane diagram which associate the two set parameters, the control characteristics as well as the component stresses of the converter can be obtained in terms of the characteristic curves. In doing so, we first derive the following equations from the state-plane diagram.

$$i_{nl}(t_o) = \left[\frac{1-M^2}{M^2}v_{nc}^2(t_o)+2[1-(\frac{\omega_{op}}{\omega_o})^2]\frac{1-M^2}{M}v_{nc}(t_o)+i_{nl}^2(t_1)\right]^{\frac{1}{2}} \quad (12)$$

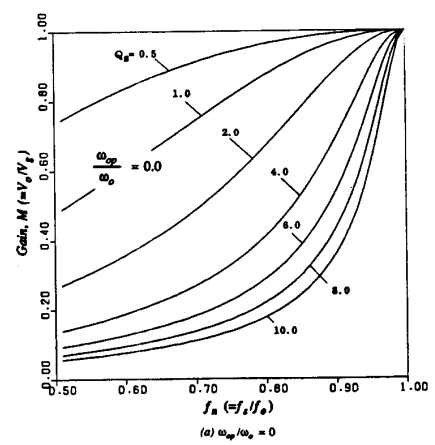
$$i_{nl}(t_1) = \frac{1}{2}(\frac{\omega_{op}}{\omega_o})^2[\gamma M+(\gamma-2\beta)] \quad (13)$$

$$\gamma = \frac{1}{MQ_s}[1-(\frac{\omega_{op}}{\omega_o})^2][2v_{nc}(t_1)-\alpha\beta(\frac{\omega_{op}}{\omega_o})^2] \quad (14)$$

$$\beta = \pi-\tan^{-1}\frac{i_{no}(t_o)}{V_{ng}-V_{no}-v_{nc}(t_o)}+\tan^{-1}\frac{i_{nl}(t_1)}{V_{ng}-V_{no}-v_{nc}(t_1)} \quad (15)$$

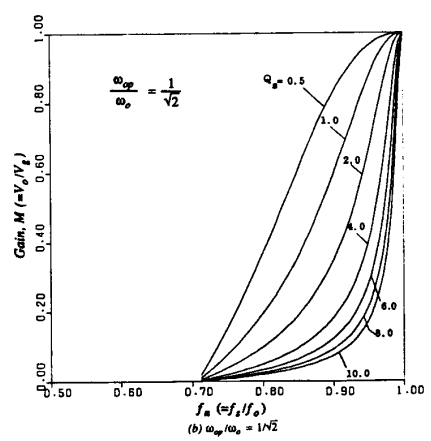
$$\alpha = \tan^{-1}\frac{v_{nc}(t_1)}{v_{nc}(t_1)-V_{ng}-V_{no}}+\tan^{-1}\frac{i_{no}(t_o)}{V_{no}-V_{ng}-v_{nc}(t_o)} \quad (16)$$

It is evident from eqs. (12)-(16) that there are six equations with nine unknowns. For given ω_{op}/ω_o, M and Q_s, the rest can be numerically solved from these equations. Thus for fixed ω_{op}/ω_o, the control characteristic curves of M vs. γ with Q_s as parameter can be plotted. Fig. 5 shows such curves with different ratio of ω_{op}/ω_o. Fig. 5(a) represents the control characteristic curves of the conventional SRC ($\omega_{op}/\omega_o = 0$). It can be seen from Fig. 5 that the control characteristics of the new converter are much improved from that of the conventional SRC while the advantage of low converter voltage gain is still maintained.



(a) $\omega_{op}/\omega_o = 0$

Fig. 5 Control characteristics M vs. f_n (=π/γ) with Q_s as parameter



(b) $\omega_{op}/\omega_o = 1/\sqrt{2}$

Fig. 5 Control characteristics M vs. f_n (=π/γ) with Q_s as parameter

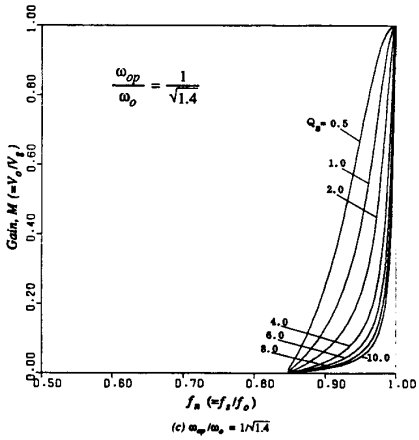


Fig. 5 Control characteristics M vs. f_n ($=\omega t/\gamma$) with Q_s as parameter

It can be shown that once the state-plane diagram is obtained, the maximum component stress can be determined. Fig. 6(a), Fig. 6(b), and Fig. 6(c) show the normalized maximum capacitor voltage v_{ncp}, inductor currents i_{nls} and i_{nlp}, respectively. These curves are useful in selection of the components in the converter design.

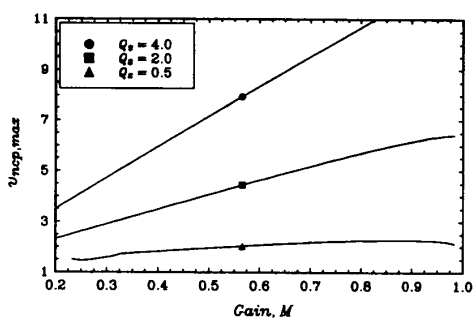


Fig. 6(a) Maximum normalized capacitor voltage $v_{cn,max}$

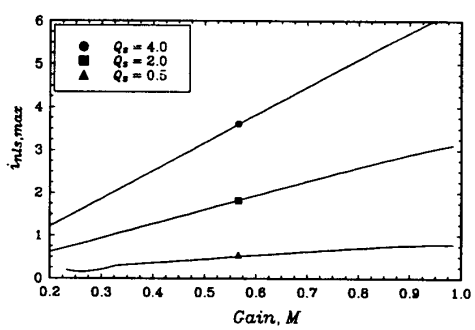


Fig. 6(b) Maximum normalized inductor current $i_{nls,max}$

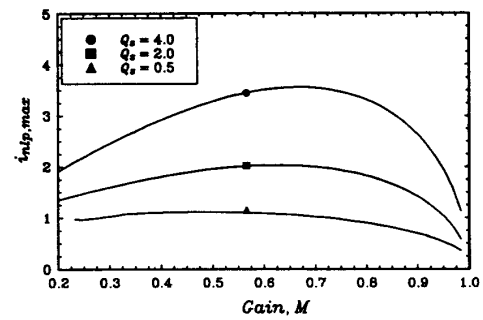


Fig. 6(c) Maximum normalized inductor current $i_{nlp,max}$

IV. DESIGN AND EXFERIMENTAL RESULTS

In this section, based on the characteristic curves given in Section III, we present a design procedure for the new converter by using a specified design example. Experimental results will also be reported.

1. Converter Design

Consider the following design specifications:

DC Supply Voltage: V_g = 115V - 185V
Output Voltages: 5V, +14V, *and* -14V
Load Currents: 5V/13A, +14V/5A, *and* -14V/1A
Load Range: 20% - 100%
Maximum Switching Frequency: 200 kHz

In this design example, the maximum output power is 150 W and the 5V output is regulated. The design of the converter is based on single output (+5V). Therefore, we use $I_{o,max}$ and $I_{o,min}$ to represent the maximum and minimum output currents at the +5V terminal, corresponding to the 100% and 20% output power, respectively. For an assumed efficiency of 80%. the values of $I_{o,max}$ and $I_{o,min}$ can be calculated from the specifications given above.

$$I_{o,max} = 37.5 \text{ A and } I_{o,min} = 3.75 \text{ A} \tag{17}$$

From the design stand-point, the maximum output power and the minimum input voltage $V_{g,min}$ represent the worst operation of the converter. Thus we choose $V_{g,min}$ = 115V and $I_{o,max}$ = 37.5A for the converter design. The following steps outline the design procedure used to determine the component values and the output transformer ratio:

(1) Choose a proper value for M and Q_s from Fig. 5(b) such that the relative change in f_{ns} is small for the specified load and line variations.

$$M = 0.8 \text{ and } Q_s = 2.0 \tag{18}$$

(2) With M and Q_s being specified, f_{ns} can be obtained from Fig. 5(b) as

$$f_{ns} \approx 0.97 \tag{19}$$

(3) The output transformer ratio N is determined from the following equation.

$$N = \frac{V_g M}{V_o + 2V_D} \approx 14 \tag{20}$$

where V_D is the rectifier voltage drop.

(4) The characteristic impedance Z_o is given by

$$Z_o = R_o Q_s = N^2 \frac{V_o + 2V_D}{I_{o,\max}} Q_s \approx 69\ \Omega \tag{21}$$

where R'_o is the equivalent load resistance seen from the primary side of the transformer.

(5) With f_s = 200 kHz and Z_o = 69Ω, the components of the tank circuit can be determined and are given by

$$C_p = \frac{f_{ns}}{2\pi f_s Z_o} \approx 0.0112\ \mu\text{F} \tag{22}$$

$$L = \frac{Z_o f_{ns}}{2\pi f_s} \approx 52\ \mu\text{H} \tag{23}$$

$$L_s = \frac{L}{1-(\omega_{op}/\omega_o)^2} \approx 104\ \mu\text{H} \tag{24}$$

$$l_p = \frac{L}{(\omega_{op}/\omega_o)^2} \approx 104\ \mu\text{H} \tag{25}$$

To predict resultant changes in switching frequency f_s under different load and line conditions, the following four cases are concerned.

(1) $I_{o,max}$ = 37.5 A and $V_{g,min}$ = 115 V
f_s = 200 kHz;

(2) $I_{o,min}$ = 3.75 A and $V_{g,min}$ = 115 V
$\Rightarrow f_s$ = 174.7 kHz;

(3) $I_{o,max}$ = 37.5 A and $V_{g,max}$ = 185 V
$\Rightarrow f_s$ = 192.3 kHz;

(4) $I_{o,min}$ = 3.75 A and $V_{g,max}$ = 185 V
$\Rightarrow f_s$ = 165.2 kHz.

Hence, for 80% load variation under the given input voltage range, the maximum required switching frequency range is Δf_s
= 34.8 kHz. The percentage change in f_s is approximately 17.4%.

2. Experimental Results

To verify the above proposed design technique, a 150 W, 200 kHz, multi-output LLC-SRC power supply has been built using the design values calculated in the design example. This power supply works from ac mains of 90 VAC to 130 VAC and has three outputs: +5V, +14V, and -14V. Fig. 7 shows the simplified circuit scheme of the converter. The actual component values of the tank circuit are $L_s = L_p$ = 104 μH and C_p = 0.011 μF. The +5V output is fed back for closed-loop control. ±14V outputs are unregulated.

Various voltage and current waveforms obtained from the experiments are shown in Fig. 8, Fig. 9, Fig. 10, and Fig. 11, which correspond to the four cases given in the previous section.

Fig. 12, Fig. 13, Fig. 14, and Fig. 15 show the cross-regulation of the ±14V outputs under different line and load conditions. It can be seen from these figures that the cross-regulation is insensitive to the line variation but very sensitive to load change. Therefore, post-regulation is needed for both ±14V outputs.

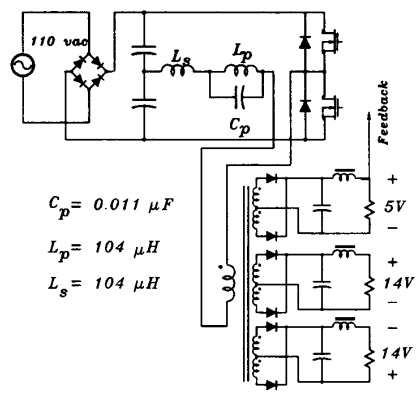
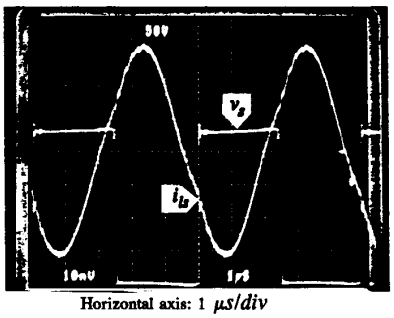


Fig. 7 Simplified circuit scheme of the implemented LLC-SRC

Fig. 16 shows the experimental verification of the control characteristics for Q_s = 0.4, 1.2, and 2.0. Table II shows the comparison between the experimental results and theoretical values under the four cases. It is evident from Fig. 16 and Table II that the experimental results are in good agreement with the calculated values.

V. CONCLUSION

Based on our previous analysis of the LLC-SRC [3], we derived a complete set of design curves for the converter. A design procedure for the converter operated in the continuous conduction mode was formulated based on these characteristic curves. As an example, a 150 W, 200 kHz, and multi-output LLC-SRC power supply has been designed and tested. Various test waveforms for different load and line conditions were presented which are in consistent to our analytical work.



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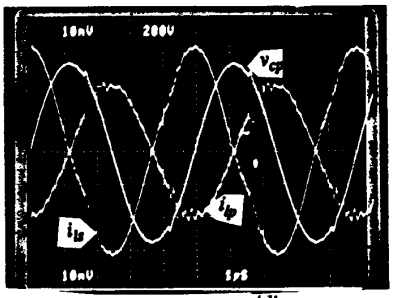
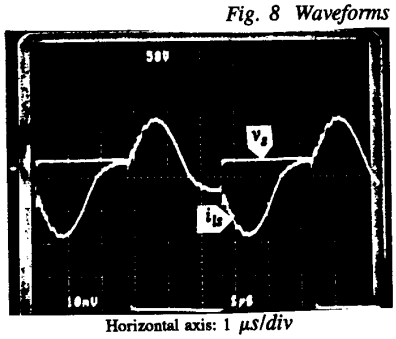
Vertical axis: v_{cp} 200 v/div and i_{ls} $1A/div$



Horizontal axis: 1 $\mu s/div$

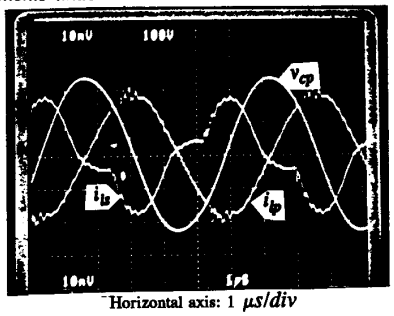
Vertical axis: v_{cp} 200 v/div, i_{ls} $1A/div$, and i_{lp} $2A/div$

Fig. 8 Waveforms obtained from the experiments under Case 1.



Horizontal axis: 1 $\mu s/div$

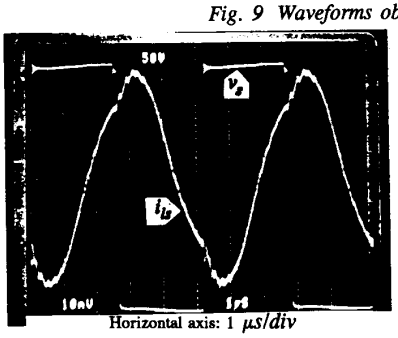
Vertical axis: v_{cp} 100 v/div and i_{ls} $0.5A/div$



Horizontal axis: 1 $\mu s/div$

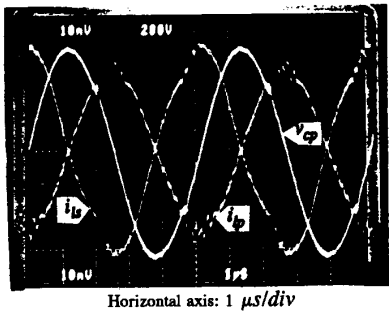
Vertical axis: v_{cp} 100 v/div, i_{ls} $0.5A/div$, and i_{lp} $1A/div$

Fig. 9 Waveforms obtained from the experiments under Case 2.



Horizontal axis: 1 $\mu s/div$

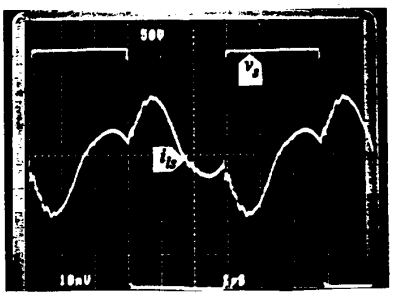
Vertical axis: v_{cp} 200 v/div and i_{ls} $1A/div$



Horizontal axis: 1 $\mu s/div$

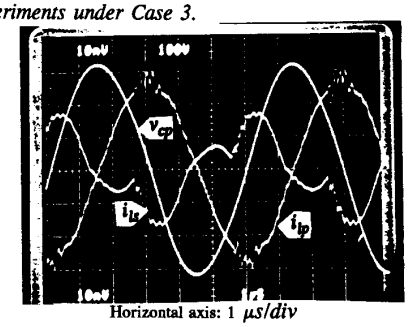
Vertical axis: v_{cp} 200 v/div, i_{ls} $1A/div$, and i_{lp} $2A/div$

Fig. 10 Waveforms obtained from the experiments under Case 3.



Horizontal axis: 1 $\mu s/div$

Vertical axis: v_{cp} 100 v/div and i_{ls} $0.5A/div$



Horizontal axis: 1 $\mu s/div$

Vertical axis: v_{cp} 100 v/div, i_{ls} $0.5A/div$, and i_{lp} $1A/div$

Fig. 11 Waveforms obtained from the experiments under Case 4.

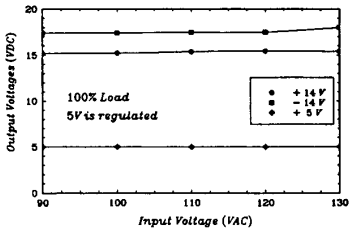


Fig. 12 Cross-regulations of ±14V outputs under line variation at 100% output load.

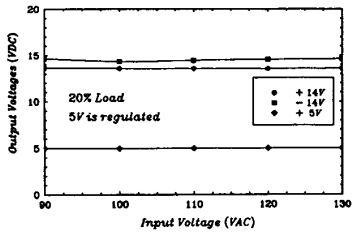


Fig. 13 Cross-regulations of ±14V outputs under line variation at 20% output load.

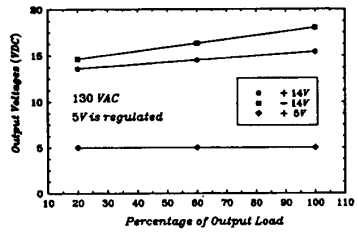


Fig. 14 Cross-regulations of ±14V outputs under load variation at 130 VAC input voltage.

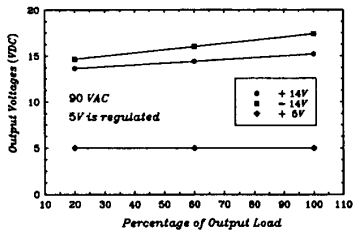


Fig. 15 Cross-regulations of ±14V outputs under load variation at 90 VAC input voltage.

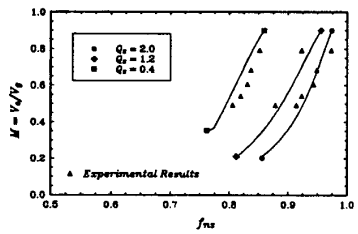


Fig. 16 Experimental verification of the control characteristics.

TABLE II

	Theoretical Values				Experimental Results			
	$v_{cp,max}$	$i_{ls,max}$	$i_{lp,max}$	f_s(kHz)	$v_{cp,max}$	$i_{ls,max}$	$i_{lp,max}$	f_s(kHz)
Case 1	655V	3.04A	4.3A	202.5	540V	3.2A	4.0A	204
Case 2	243V	0.97A	1.51A	176.9	240V	0.8A	1.8A	179
Case 3	749V	4.3A	5.2A	194.5	640V	3.4A	4.5A	192
Case 4	341V	0.99A	2.9A	167.2	320V	0.7A	2.7A	169

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